

# Safe Adaptive Quantization for Ultra-Low-Bitrate Generative Image Compression

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**Abstract**—Modern image compression methods are built from neural networks: instead of hand-designed transforms, they learn to turn an image into a compact set of features and to model the statistics of those features so they can be stored in few bits. At very low bitrates, optimizing for pixel accuracy makes reconstructions look blurry and unnatural, so recent methods attach a generative decoder that *synthesizes* realistic detail. This creates an opportunity: if the decoder can plausibly invent texture on its own, the encoder need not spend bits describing detail the decoder could reproduce anyway. We turn this into a method. On top of an existing, *frozen* generative compression model, we add a small module that decides, for each region of the image, whether its fine detail can be dropped without visibly hurting quality, and quantizes that region more coarsely when it can. The decision is reconstructed by the decoder from information it already receives, so nothing extra is transmitted, and the module reduces to the original model when switched off. Trained with a criterion that permits coarsening only where it saves bits without harming appearance, our method—Safe Adaptive Quantization (SAQ)—reduces the rate at equal perceptual quality by about 10% in DISTS and 6% in FID on the CLIC2020 and DIV2K benchmarks, without retraining the underlying model. An analysis confirms that the learned module preserves edges and texture while economizing on flat, predictable regions.

**Index Terms**—Learned image compression, generative compression, perceptual quality, adaptive quantization, ultra-low bitrate.

## I. INTRODUCTION

Image compression decides which information in a picture to keep and which to discard. Modern systems learn this decision from data: a neural network maps an image to a compact set of features, a second network models the probabilities of those features so they can be written with few bits, and a decoder network reconstructs the image. This paradigm, learned image compression (LIC), has grown from transform autoencoders with hyperpriors [1], [2] into strong systems whose entropy models exploit autoregressive, channel-wise, and transformer-based context [3]–[7]. These systems are tuned for rate–distortion, but at very low bitrates a distortion objective rewards blurry, averaged reconstructions that look unnatural to a human observer. This mismatch is formalized by the rate–distortion–perception trade-off [8], [9] and has driven a shift toward *generative* compression, where the decoder is trained to produce realistic detail at a small rate cost [10]–[12].

A generative decoder changes what the bitstream must carry. Because the decoder can synthesize plausible high-frequency content from a compact code, it no longer needs

every detail to be transmitted explicitly. Vector-quantized and diffusion decoders push this idea to extreme rates: Generative Latent Coding (GLC) [13], [14] performs transform coding in the latent space of a generative VQ autoencoder [15] and reports high realism below 0.05 bpp, and diffusion codecs reconstruct recognizable images at a few thousandths of a bit per pixel [16], [17].

This raises a simple question that, to our knowledge, is rarely asked directly: *if the decoder can regenerate part of the detail, why spend the same coding precision everywhere?* Spatially adaptive bit allocation is a classic idea, but in LIC it is usually realized either by transmitting a content importance map [18]—which adds to the rate—or by retraining the codec for each operating point [19]. Neither fits a setting where we want to exploit the recovery ability of an *existing, frozen* generative codec without paying for extra side information.

We address exactly this setting. Our method, Safe Adaptive Quantization (SAQ), learns *where* the latent can be quantized more coarsely without perceptual damage, and—crucially—makes that decision recomputable at the decoder so it costs no bits. SAQ is a small module attached to a frozen GLC codec. From the information the decoder already has (the transmitted hyper code and the partially decoded latent), it predicts a per-region gate  $\gamma \geq 1$  that enlarges the quantization step. Because  $\gamma \geq 1$ , SAQ never codes the latent more finely than the base codec, and a zero initialization makes it identical to GLC, so any gain comes purely from coarsening where it is safe. To decide what is safe, SAQ is trained with a benefit–cost teacher that rewards a larger gate only where coarsening saves rate while keeping the reconstruction close to the original.

Our contributions are:

- We show that a frozen state-of-the-art generative codec spends latent precision on detail its own decoder can regenerate, and that this precision can be reduced *spatially* and *without side information*.
- We propose SAQ, a lightweight decoder-recomputable quantization gate trained with a benefit–cost (safe-coarsening) teacher; it adds no side channel, is monotone ( $\gamma \geq 1$ ), and reproduces the base codec at initialization.
- Without retraining the base codec, SAQ reduces the Bjøntegaard rate [20] by about 10% (DISTS) and 6% (FID) on CLIC2020 and DIV2K, and we analyze where and why the savings arise.

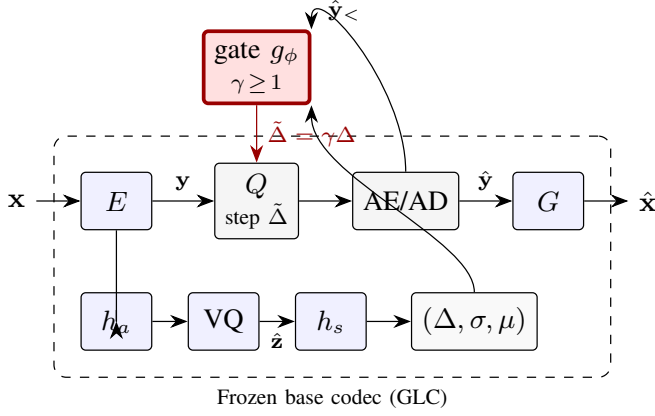


Fig. 1. SAQ on a frozen generative codec. The hyper branch produces a content-adaptive quantization step  $\Delta$  (with scale  $\sigma$  and mean  $\mu$ ). A small gate  $g_\phi$  predicts  $\gamma \geq 1$  from information already available at the decoder—the transmitted hyper code and the partially decoded latent  $\hat{y}_<$ —and enlarges the step to  $\hat{\Delta} = \gamma\Delta$ . The gate is recomputed at the decoder, so no map is transmitted;  $\gamma \equiv 1$  recovers the base codec.

## II. RELATED WORK

**Learned image compression.** End-to-end transform coding with hyperpriors [1], [2] and increasingly expressive entropy models [3]–[7] defines the rate–distortion state of the art; we build on the open CompressAI-style tooling [21]. These works improve  $p(y)$  or the transform; we instead change *how coarsely* parts of  $y$  are quantized for a fixed codec.

**Perceptual and generative compression.** Optimizing for realism rather than distortion improves low-rate visual quality [9]. GAN-based codecs [10]–[12] and VQ-/diffusion-based generative codecs [13], [14], [16], [17] reach ultra-low bitrates by synthesizing detail. GLC [13] is our frozen base codec. SAQ is orthogonal: it is a bit-allocation layer that can sit on such a codec to spend fewer bits where the generator already suffices.

**Adaptive bit allocation.** Content-weighted coding transmits an importance map to allocate bits [18], and conditional autoencoders enable variable-rate coding [19]. Unlike these, SAQ sends no map and does not retrain the codec; its allocation is recomputed at the decoder, and it is supervised to coarsen only where it is perceptually safe.

## III. METHOD

### A. Background: latent coding in a generative codec

We build on GLC [13]. An encoder maps the image  $x$  to a latent  $y = E(x)$  in the space of a generative VQ autoencoder [15]; a generator reconstructs  $\hat{x} = G(\hat{y})$ . A hyper-analysis transform produces a summary  $z$  that is vector-quantized and transmitted as codebook indices; a hyper-synthesis transform turns the dequantized  $\hat{z}$  into a content-adaptive quantization step  $\Delta$ , a scale  $\sigma$  and a mean  $\mu$  for the latent. Each latent element is coded as

$$r = \left\lfloor \frac{y}{\Delta} - \mu \right\rfloor, \quad \hat{y} = \Delta(r + \mu), \quad (1)$$

where the residual  $r$  is entropy-coded under a Gaussian model with scale  $\sigma$ , and  $\lfloor \cdot \rfloor$  is rounding. The latent is decoded in four spatial passes, so  $\sigma$  and  $\mu$  for later positions are conditioned on the already-decoded latent. The side information ( $\hat{z}$  indices) is small and, as we confirm below, unchanged by our method.

### B. Safe adaptive quantization

The step  $\Delta$  in (1) is already content-adaptive, but it is optimized for the codec’s own rate–distortion; it does not account for the fact that the generator  $G$  can regenerate some detail. SAQ inserts a multiplicative gate. Let  $c$  denote the information available at the decoder when a latent position is coded: the hyper-derived parameters  $(\Delta, \sigma, \mu)$  and the partially decoded latent  $\hat{y}_<$ . A small network  $g_\phi$  predicts

$$\gamma = 1 + (\gamma_{\max} - 1) \psi(g_\phi(c)) \in [1, \gamma_{\max}], \quad (2)$$

where  $\psi(\cdot) \in [0, 1]$  is a smooth squashing function and  $\gamma_{\max}$  caps the coarsening. The quantization step is replaced by  $\hat{\Delta} = \gamma\Delta$  in (1). Three properties make this usable on a frozen codec without side information (Fig. 1):

- (i) *No side channel.*  $c$  contains only signals the decoder also has, so  $\gamma$  is recomputed identically at decode time; nothing extra is transmitted.
- (ii) *Monotonicity.* Since  $\gamma \geq 1$ , the step never shrinks: SAQ can only drop precision relative to GLC, never request more.
- (iii) *Exact fallback.* The final layer of  $g_\phi$  is zero-initialized, so  $\gamma \equiv 1$  at the start and SAQ reproduces GLC exactly; any improvement is attributable solely to learned coarsening.

A complementary head, sharing the same decoder-available context  $c$  and the same zero-initialization, refines the entropy parameters  $(\sigma, \mu)$  used to code the residual. It carries no side information and tightens the entropy model; the gate is the component that reallocates rate.

### C. Safe-coarsening objective

A plain rate term would push  $\gamma$  up everywhere and blur the image. The key is to coarsen *only where it is safe*. During training we therefore form a benefit–cost target from the frozen base reconstruction. For a candidate coarsening we measure, per region, the increase in fidelity error ( $\Delta L_1$ ,  $\Delta \text{LPIPS}$  relative to GLC) and the rate it saves ( $\Delta R$ ), and define a safety map

$$s = \sigma \left( \frac{1}{\tau} [w_R \widetilde{\Delta R} - w_1 \widetilde{\Delta L_1} - w_p \widetilde{\Delta \text{LPIPS}}] \right), \quad (3)$$

where  $\widetilde{(\cdot)}$  denotes per-image standardization and  $\sigma$  is the logistic function. A region is “safe” ( $s \rightarrow 1$ ) when coarsening saves bits with little perceptual or fidelity damage. The full objective is

$$\mathcal{L} = R + \lambda_p \text{LPIPS}(x, \hat{x}) + \lambda_s \text{BCE}(\psi(g_\phi(c)), s) + \lambda_\gamma (\bar{\gamma} - \tau_\gamma)^2, \quad (4)$$

where  $R$  is the estimated rate, the third term aligns the gate with the safety map, and the last term sets the operating point through a target mean gate  $\tau_\gamma$  (a single scalar that trades rate against quality). The base codec stays frozen, and a learned rate embedding lets one SAQ module serve the four operating

TABLE I  
BD-RATE (%) VS. THE FROZEN GLC CODEC [13]. NEGATIVE IS BETTER (FEWER BITS AT EQUAL QUALITY). DISTS AND FID ARE THE PRIMARY PERCEPTUAL METRICS. “W/O SAFE TEACHER” TRAINS THE SAME GATE WITHOUT THE BENEFIT–COST TARGET OF (3).

| Dataset  | Method                | DISTS  | LPIPS | FID   | KID   |
|----------|-----------------------|--------|-------|-------|-------|
| CLIC2020 | SAQ, w/o safe teacher | −8.37  | +0.68 | −5.50 | −4.39 |
|          | <b>SAQ (full)</b>     | −9.76  | +0.05 | −6.42 | −4.65 |
| DIV2K    | SAQ, w/o safe teacher | −9.20  | −0.05 | −5.13 | −1.23 |
|          | <b>SAQ (full)</b>     | −10.37 | −0.49 | −5.64 | −4.98 |

points of GLC. We use LPIPS [22] as the perceptual term; DISTS and FID are reserved for evaluation.

#### IV. EXPERIMENTS

##### A. Setup

**Base codec and training.** We attach SAQ to the official GLC image codec [13] and keep it frozen; only the SAQ heads (about 6% of the backbone parameters) are trained, on OpenImages [23]. We report the four GLC operating points.

**Datasets.** Our rate–perception evaluation uses the CLIC2020 test set (428 professional and mobile images) [24] and the DIV2K validation set (100 images) [25]; we use Kodak (24 images) [26] for the gate analysis and qualitative examples. All images are coded at full resolution.

**Metrics.** Following the perceptual-compression literature, our primary metrics are DISTS [27] and FID [28], which correlate with low-rate visual quality; we also report LPIPS [22] and KID [29], and PSNR/MS-SSIM as distortion diagnostics. FID/KID use  $256 \times 256$  patches. We summarize rate–quality with Bjøntegaard delta rate [20] (BD-rate), where a negative value means fewer bits at equal quality.

##### B. Rate–perception performance

Table I reports BD-rate against the frozen GLC codec, and Fig. 2 shows the corresponding curves. SAQ moves the rate–perception curve left on both datasets: it needs 9.8% and 10.4% fewer bits at equal DISTS, and 6.4% and 5.6% fewer bits at equal FID, on CLIC2020 and DIV2K. KID improves consistently (−4.7%/−5.0%). LPIPS is essentially unchanged (Fig. 2, right), which we attribute to its sensitivity to the slight smoothing introduced by coarsening; we therefore do not claim a gain on every metric and keep LPIPS as a neutral diagnostic. Because the gate sends no side information and leaves the hyper code untouched, the side-information rate is identical to GLC (0.0035 bpp on both sets); all savings come from the latent. Fig. 3 shows the effect visually: at the top operating point SAQ matches GLC at the same DISTS while using 8–9% fewer bits.

##### C. Effect of the safe-coarsening teacher

The lower rows of Table I ablate the benefit–cost target of (3). Training the same gate without it—i.e. letting the rate term distribute coarsening on its own—already helps, but the safety teacher improves the DISTS BD-rate from −8.4% to

−9.8% on CLIC2020 and from −9.2% to −10.4% on DIV2K, with matching FID and KID gains. Telling the gate *where* coarsening is safe, rather than only *how much* to coarsen on average, is what turns a generic rate knob into a useful spatial allocation.

##### D. What does the gate learn?

Fig. 4 visualizes  $\gamma$  at the highest rate. Coarsening concentrates on flat, predictable regions (walls, sky, road) and backs off on edges, text, and fine texture. We quantify this on Kodak by correlating  $\gamma$  with local statistics of the image and of the base reconstruction. Averaged over the 24 images, the gate is negatively correlated with the base reconstruction error (−0.51), local texture variance (−0.47), and gradient magnitude (−0.52): it raises precision-saving exactly where GLC is already accurate and the content is smooth. Regions in the top gate quartile have less than half the reconstruction error of the bottom quartile (0.030 vs. 0.074). This is the intended mechanism: spend bits on residual the generator cannot infer, and coarsen the rest.

##### E. Complexity

SAQ keeps the GLC backbone frozen and adds only lightweight heads ( $\approx 6\%$  of the backbone parameters; the gate alone is  $< 3\%$ ). The gate is a small convolutional network evaluated alongside the existing entropy model, so it adds negligible cost to encoding and decoding and requires no change to the bitstream format.

#### V. CONCLUSION

We presented Safe Adaptive Quantization, a side-information-free way to reallocate coding precision in a frozen generative image codec. SAQ learns a decoder-recomputable gate that coarsens the latent only where coarsening is perceptually safe, supervised by a benefit–cost teacher. It reduces the Bjøntegaard rate by about 10% in DISTS and 6% in FID on CLIC2020 and DIV2K, with no change to the base codec and no extra bits, and the learned gate demonstrably targets predictable, low-error regions. The principle—reserve bits for what the generator cannot regenerate—is general, and SAQ realizes it as a drop-in layer.

**Scope and future work.** We study the ultra-low-bitrate regime of one base codec; LPIPS is left essentially unchanged, reflecting the perception–distortion nature of coarsening. Promising directions are learning to *synthesize* the omitted residual at the decoder rather than only dropping it, and applying the same side-information-free gate to other generative and diffusion codecs.

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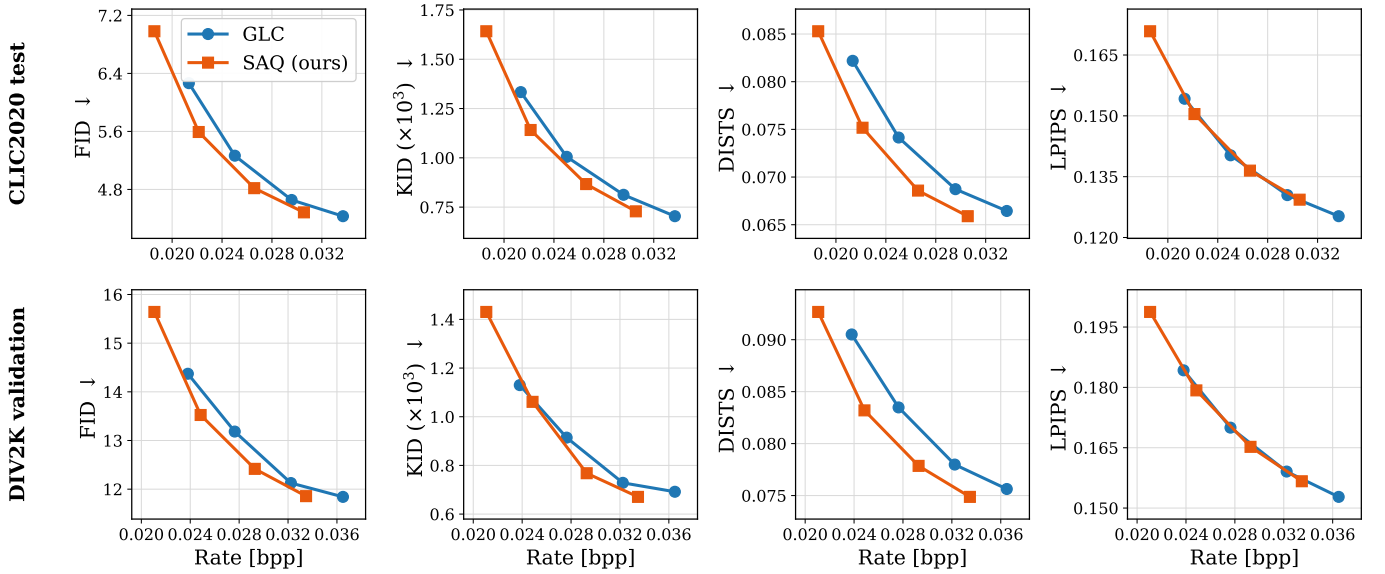


Fig. 2. Rate–quality curves over the four operating points, for two distribution metrics (FID, KID) and two full-reference perceptual metrics (DISTS, LPIPS); lower-left is better. At a fixed quality, SAQ (orange) lies to the left of GLC (blue), i.e. it needs fewer bits for the same FID, KID and DISTS on both datasets. LPIPS, which is sensitive to mild smoothing, stays on the GLC curve. KID is scaled by  $10^3$ .

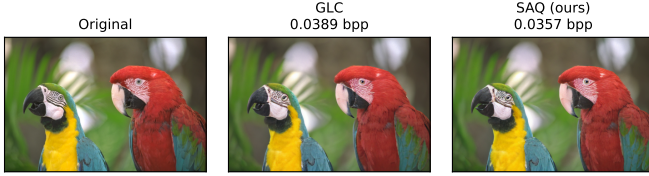


Fig. 3. Reconstruction at the top operating point (Kodak, kodim23). SAQ matches the frozen GLC codec in DISTS (0.068 for both) while spending 8.3% fewer bits; the side information is unchanged. Best viewed zoomed in.

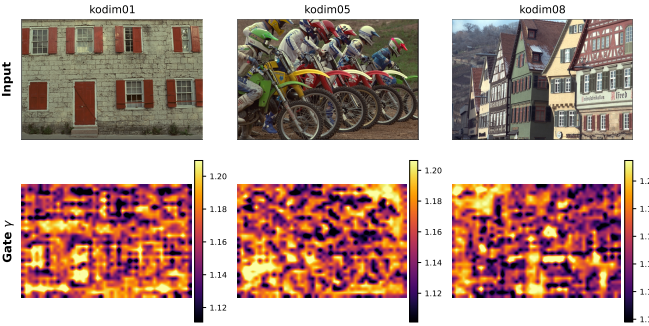


Fig. 4. Learned gate  $\gamma$  (higher = coarser) at the highest rate. The gate enlarges the quantization step on flat regions (walls, road) and protects edges, windows, and texture.

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